Appendix D

Numerical implementation of the collision operator in Gysela

In this appendix, the simplified expression of the Lorentz-type operator which is used in GYSELA is detailed. The expression of this collision operator (including the perpendicular direction, which is not yet implemented in GYSELA) is

$$\mathcal{C}(\bar{F}_s) = \frac{1}{B^*_{\|s}}\partial_{v_{G\|}}\left\{B^*_{\|s}\mathcal{D}_{\|s}\bar{F}_{M_s}\partial_{v_{G\|}}\left(\frac{\bar{F}_s}{\bar{F}_{M_s}}\right)\right\} + \frac{1}{B^*_{\|s}}\partial_{\mu}\left\{B^*_{\|s}\mathcal{D}_{\perp s}\bar{F}_{M_s}\frac{1}{B^2}\partial_{\mu}\left(\frac{\bar{F}_s}{\bar{F}_{M_s}}\right)\right\} \tag{D.1}$$

where $B^*_{\|s}(r,\theta,v_{G\|}) = B(r,\theta) + v_{G\|}\mathbf{b}\cdot\nabla\times\mathbf{b}$ is the Jacobian of the guiding-center coordinates, and $\bar{F}_{M_s}$ is the following shifted Maxwellian distribution

$$\bar{F}_{M_s} = \frac{n_{s_0}}{\left(2\pi T_{s,\mathrm{coll}}/m_s\right)^{3/2}}\exp\left(-\frac{m_s\left(v_{G\|}-V_{\|s,\mathrm{coll}}\right)^2}{2T_{s,\mathrm{coll}}} - \frac{\mu_s B}{T_{s,\mathrm{coll}}}\right) \tag{D.2}$$

with the mean temperature $T_{s,\mathrm{coll}} = T_{s,\mathrm{coll}}(r,\theta,\varphi)$ and the mean velocity $V_{\|s,\mathrm{coll}} = V_{\|s,\mathrm{coll}}(r,\theta,\varphi)$. $\bar{F}_{M_s}$ is such that $\mathcal{C}(\bar{F}_s) = 0$. The collision term $\mathcal{D}_{\|s}$ is defined by the equations (2.23)-(2.27). The expression of $T_{s,\mathrm{coll}}$ and $V_{\|s,\mathrm{coll}}$ are constrained such that the collision operator is momentum and energy preserving. These calculations and expressions are detailed in a first paragraph D.1 while the numerical implementation based on a semi-implicit second order Crank-Nicolson scheme is described in the second one D.2.

D.1 Expressions of the mean temperature $T_{s,\mathrm{coll}}$ and mean velocity $V_{\|s,\mathrm{coll}}$ for the collision operator

While this simplified collision operator obviously conserves the number of particles, the profiles $T_{s,\mathrm{coll}}(r,\theta,\varphi)$ and $V_{\|s,\mathrm{coll}}(r,\theta,\varphi)$ must be chosen so that the operator is also compatible with the conservations of momentum and energy

$$\int \mathcal{J}_{\mathrm{v}}\,\mathrm{d}\mu_s\,\mathrm{d}v_{G\|}\; m_s v_{G\|}\mathcal{C}(\bar{F}_s) = 0 \tag{D.3}$$

$$\int \mathcal{J}_{\mathrm{v}}\,\mathrm{d}\mu_s\,\mathrm{d}v_{G\|}\left(\mu_s B + \frac{1}{2}m_s v_{G\|}^2\right)\mathcal{C}(\bar{F}_s) = 0 \tag{D.4}$$

where $\mathcal{J}_{\mathrm{v}} = 2\pi B^*_{\|s}/m_s$ is the jacobian in velocity space. Let us first consider only the contribution of collisions in the parallel direction. We use the expression of the collision

operator (D.1) and integrate Eq. (D.3) by parts twice in the variable $v_{G\parallel}$

$$\begin{aligned}
(\mathrm{D.3})_\parallel &= \int \frac{2\pi B^*_{\parallel s}}{m_s}\,\mathrm{d}\mu_s\,\mathrm{d}v_{G\parallel}\,m_s v_{G\parallel}\frac{1}{B^*_{\parallel s}}\partial_{v_{G\parallel}}\left\{B^*_{\parallel s}\mathcal{D}_{\parallel s}\bar{F}_{M_s}\partial_{v_{G\parallel}}\left(\frac{\bar{F}_s}{\bar{F}_{M_s}}\right)\right\}\\
&= -2\pi\int \mathrm{d}\mu_s\,\mathrm{d}v_{G\parallel}\,B^*_{\parallel s}\mathcal{D}_{\parallel s}\bar{F}_{M_s}\partial_{v_{G\parallel}}\left(\frac{\bar{F}_s}{\bar{F}_{M_s}}\right)\\
&= 2\pi\int \mathrm{d}\mu_s\,\mathrm{d}v_{G\parallel}\,\frac{\bar{F}_s}{\bar{F}_{M_s}}\partial_{v_{G\parallel}}(B^*_{\parallel s}\mathcal{D}_{\parallel s}\bar{F}_{M_s})\\
&= 2\pi\int \mathrm{d}\mu_s\,\mathrm{d}v_{G\parallel}\,\left\{\frac{\bar{F}_s}{\bar{F}_{M_s}}B^*_{\parallel s}\mathcal{D}_{\parallel s}\partial_{v_{G\parallel}}\bar{F}_{M_s}+\bar{F}_s\partial_{v_{G\parallel}}(B^*_{\parallel s}\mathcal{D}_{\parallel s})\right\}
\end{aligned}$$

Using the fact that $\partial_{v_{G\parallel}}\bar{F}_{M_s}=-\bar{F}_{M_s}\frac{m_s(v_{G\parallel}-V_{\parallel s,\mathrm{coll}})}{T_{s,\mathrm{coll}}}$, we obtain

$$\begin{aligned}
(\mathrm{D.3})_\parallel &= \frac{2\pi}{T_{s,\mathrm{coll}}}\int B^*_{\parallel s}\,\mathrm{d}\mu_s\,\mathrm{d}v_{G\parallel}\left\{\frac{T_{s,\mathrm{coll}}\bar{F}_s}{B^*_{\parallel s}}\partial_{v_{G\parallel}}(B^*_{\parallel s}\mathcal{D}_{\parallel s})-\mathcal{D}_{\parallel s}\bar{F}_s m_s(v_{G\parallel}-V_{\parallel s,\mathrm{coll}})\right\}\\
&= \frac{m_s}{T_{s,\mathrm{coll}}}\left\{V_{\parallel s,\mathrm{coll}}\left\langle m_s\mathcal{D}_{\parallel s}\right\rangle-\left\langle m_s\mathcal{D}_{\parallel s}v_{G\parallel}\right\rangle+T_{s,\mathrm{coll}}\left\langle\frac{1}{B^*_{\parallel s}}\partial_{v_{G\parallel}}(B^*_{\parallel s}\mathcal{D}_{\parallel s})\right\rangle\right\} \qquad (\mathrm{D.5})
\end{aligned}$$

where

$$\langle ...\rangle = \int \mathcal{J}_\mathrm{v}\,\mathrm{d}\mu_s\,\mathrm{d}v_{G\parallel}\,\bar{F}_s... \qquad (\mathrm{D.6})$$

We perform similar operations on the contribution to Eq. (D.4) of collisions in the parallel direction:

$$\begin{aligned}
(\mathrm{D.4})_\parallel &= \int \mathcal{J}_\mathrm{v}\,\mathrm{d}\mu_s\,\mathrm{d}v_{G\parallel}(\mu_s B+\frac{1}{2}m_s v^2_{G\parallel})\frac{1}{B^*_{\parallel s}}\partial_{v_{G\parallel}}\left\{B^*_{\parallel s}\mathcal{D}_{\parallel s}\bar{F}_{M_s}\partial_{v_{G\parallel}}\left(\frac{\bar{F}_s}{\bar{F}_{M_s}}\right)\right\}\\
&= -2\pi\int \mathrm{d}\mu_s\,\mathrm{d}v_{G\parallel}\,B^*_{\parallel s}\mathcal{D}_{\parallel s}\bar{F}_{M_s}v_{G\parallel}\partial_{v_{G\parallel}}\left(\frac{\bar{F}_s}{\bar{F}_{M_s}}\right)\\
&= 2\pi\int \mathrm{d}\mu_s\,\mathrm{d}v_{G\parallel}\frac{\bar{F}_s}{\bar{F}_{M_s}}\partial_{v_{G\parallel}}(B^*_{\parallel s}\mathcal{D}_{\parallel s}v_{G\parallel}\bar{F}_{M_s})\\
&= \frac{2\pi}{T_{s,\mathrm{coll}}}\int B^*_{\parallel s}\,\mathrm{d}\mu_s\,\mathrm{d}v_{G\parallel}\left\{-v_{G\parallel}\mathcal{D}_{\parallel s}\bar{F}_s m_s(v_{G\parallel}-V_{\parallel s,\mathrm{coll}})+\frac{T_{s,\mathrm{coll}}}{B^*_{\parallel s}}\bar{F}_s\partial_{v_{G\parallel}}(B^*_{\parallel s}v_{G\parallel}\mathcal{D}_{\parallel s})\right\}\\
&= \frac{m_s}{T_{s,\mathrm{coll}}}\left\{V_{\parallel s,\mathrm{coll}}\left\langle m_s\mathcal{D}_{\parallel s}v_{G\parallel}\right\rangle-\left\langle m_s\mathcal{D}_{\parallel s}v^2_{G\parallel}\right\rangle+T_{s,\mathrm{coll}}\left\langle\frac{1}{B^*_{\parallel s}}\partial_{v_{G\parallel}}(B^*_{\parallel s}\mathcal{D}_{\parallel s}v_{G\parallel})\right\rangle\right\} \qquad (\mathrm{D.7})
\end{aligned}$$

For collisions in the perpendicular direction, Eq. (D.3) is trivially verified. We perform two integrations by parts in the variable μ_s for Eq. (D.4)

$$\begin{aligned}
(\mathrm{D.4})_\perp &= \int \mathcal{J}_\mathrm{v}\,\mathrm{d}v_{G\parallel}\,\mathrm{d}\mu_s\,(\mu_s B+\frac{1}{2}m_s v^2_{G\parallel})\frac{1}{B^*_{\parallel s}}\partial_\mu\left\{B^*_{\parallel s}\mathcal{D}_{\perp s}\bar{F}_{M_s}\frac{1}{B^2}\partial_\mu\left(\frac{\bar{F}_s}{\bar{F}_{M_s}}\right)\right\}\\
&= -\frac{2\pi}{m_s}\int \mathrm{d}v_{G\parallel}\,\mathrm{d}\mu_s\,\frac{B^*_{\parallel s}}{B}\mathcal{D}_{\perp s}\bar{F}_{M_s}\partial_\mu\left(\frac{\bar{F}_s}{\bar{F}_{M_s}}\right)\\
&= \frac{2\pi}{m_s}\int \mathrm{d}v_{G\parallel}\,\mathrm{d}\mu_s\frac{\bar{F}_s}{\bar{F}_{M_s}}\frac{1}{B}\partial_\mu(B^*_{\parallel s}\mathcal{D}_{\perp s}\bar{F}_{M_s})
\end{aligned}$$

Using the fact that $\partial_\mu \bar{F}_{M_s} = -\bar{F}_{M_s} \frac{B}{T_{s,\text{coll}}}$, we obtain

$$\begin{aligned}(\text{D.4})_\perp &= \frac{2\pi}{m_s} \int \mathrm{d}v_{G\|}\, \mathrm{d}\mu_s \left\{ \frac{\bar{F}_s}{B} \partial_\mu (B^*_{\|s} \mathcal{D}_{\perp s}) - B^*_{\|s} \mathcal{D}_{\perp s} \frac{1}{T_{s,\text{coll}}} \bar{F}_s \right\} \\ &= \frac{1}{T_{s,\text{coll}}} \left\{ \left\langle \frac{1}{B B^*_{\|s}} \partial_\mu (B^*_{\|s} \mathcal{D}_{\perp s}) \right\rangle T_{s,\text{coll}} - \langle \mathcal{D}_{\perp s} \rangle \right\} \end{aligned} \tag{D.8}$$

Using Equations (D.5), (D.7) and (D.8), the conservation equations (D.3) and (D.4) form a linear system in $T_{s,\text{coll}}(r,\theta,\varphi)$ and $V_{\|s,\text{coll}}(r,\theta,\varphi)$ as follows

$$\begin{aligned} V_{\|s,\text{coll}} \left\langle m_s \mathcal{D}_{\|s} \right\rangle + T_{s,\text{coll}} \left\langle \frac{1}{B^*_{\|s}} \partial_{v_{G\|}} (B^*_{\|s} \mathcal{D}_{\|s}) \right\rangle &= \left\langle m_s \mathcal{D}_{\|s} v_{G\|} \right\rangle \\ V_{\|s,\text{coll}} \left\langle m_s^2 \mathcal{D}_{\|s} v_{G\|} \right\rangle + T_{s,\text{coll}} \left\langle \frac{m_s}{B^*_{\|s}} \partial_{v_{G\|}} (B^*_{\|s} \mathcal{D}_{\|s} v_{G\|}) + \frac{1}{B B^*_{\|s}} \partial_\mu (B^*_{\|s} \mathcal{D}_{\perp s}) \right\rangle &= \left\langle m_s^2 \mathcal{D}_{\|s} v_{G\|}^2 + \mathcal{D}_{\perp s} \right\rangle \end{aligned}$$

Solving this system, we find that the conservation constraints are verified if the profiles $V_{\|s,\text{coll}}(r,\theta,\varphi)$ and $T_{s,\text{coll}}(r,\theta,\varphi)$ for the collision operator are defined as follows

$$\begin{aligned} m_s P V_{\|s,\text{coll}} &= \left\langle \frac{m_s}{B^*_{\|s}} \partial_{v_{G\|}} (B^*_{\|s} \mathcal{D}_{\|s} v_{G\|}) + \frac{1}{B B^*_{\|s}} \partial_\mu (B^*_{\|s} \mathcal{D}_{\perp s}) \right\rangle \left\langle m_s \mathcal{D}_{\|s} v_{G\|} \right\rangle \\ &\quad - \left\langle \frac{1}{B^*_{\|s}} \partial_{v_{G\|}} (B^*_{\|s} \mathcal{D}_{\|s}) \right\rangle \left\langle m_s^2 \mathcal{D}_{\|s} v_{G\|}^2 + \mathcal{D}_{\perp s} \right\rangle \\ P T_{s,\text{coll}} &= \left\langle \mathcal{D}_{\|s} \right\rangle \left\langle m_s^2 \mathcal{D}_{\|s} v_{G\|}^2 + \mathcal{D}_{\perp s} \right\rangle - \left\langle m_s \mathcal{D}_{\|s} v_{G\|} \right\rangle^2 \end{aligned}$$

where

$$P = \left\langle \mathcal{D}_{\|s} \right\rangle \left\langle \frac{m_s}{B^*_{\|s}} \partial_{v_{G\|}} (B^*_{\|s} \mathcal{D}_{\|s} v_{G\|}) + \frac{1}{B B^*_{\|s}} \partial_\mu (B^*_{\|s} \mathcal{D}_{\perp s}) \right\rangle - \left\langle m_s \mathcal{D}_{\|s} v_{G\|} \right\rangle \left\langle \frac{1}{B^*_{\|s}} \partial_{v_{G\|}} (B^*_{\|s} \mathcal{D}_{\|s}) \right\rangle$$

We recall that

$$\langle ... \rangle = \int \mathcal{J}_\text{v}\, \mathrm{d}\mu_s\, \mathrm{d}v_{G\|} \bar{F}_s ... \quad \text{with} \quad \mathcal{J}_\text{v} = 2\pi B^*_{\|s}/m_s$$

Let us remind that only collisions in the parallel direction are taken into account at the moment in the code, i.e $\mathcal{D}_{\perp s} = 0$ in above expressions. Then, let us define the 5 following integrals:

$$\begin{aligned} &I_0 = \langle \mathcal{D}_{\|s} \rangle \quad ; \quad I_1 = \langle m_s \mathcal{D}_{\|s} v_{G\|} \rangle \quad ; \quad I_2 = \langle m_s^2 \mathcal{D}_{\|s} v_{G\|}^2 \rangle \\ &I_3 = \langle \frac{1}{B^*_{\|s}} \partial_{v_{G\|}} \left(B^*_{\|s}\, \mathcal{D}_{\|s} \right) \rangle \quad \text{and} \quad I_4 = \langle \frac{m_s}{B^*_{\|s}} \partial_{v_{G\|}} \left(B^*_{\|s}\, v_{G\|}\, \mathcal{D}_{\|s} \right) \rangle \end{aligned}$$

Therefore the mean velocity and mean temperature can be simply expressed as:

$$\begin{aligned} m_s V_{\|s,\text{coll}} &= P^{-1} \left(I_4 \times I_1 - I_2 \times I_3 \right) \\ T_{s,\text{coll}} &= P^{-1} \left(I_0 \times I_2 - I_1^2 \right) \\ P &= I_0 \times I_4 - I_1 \times I_3 \end{aligned}$$

D.2 Crank-Nicolson scheme for collision operator solving

In the following, the semi-implicit second-order Crank-Nicolson scheme which is implemented in GYSELA to take into account the collisional effects as

$$\partial_t \bar{F}_s = \mathcal{C}_{ss}(\bar{F}_s) = \frac{1}{B^*_{\|s}} \partial_{v_{G\|}} \left\{ B^*_{\|s} \mathcal{D} \partial_{v_{G\|}} \bar{F}_s - B^*_{\|s} \mathcal{V} \bar{F}_s \right\} \tag{D.9}$$

is detailed. The diffusion term $\mathcal{D}$ is defined as $\mathcal{D} = \mathcal{D}_{\|s}(r,v) = \mathcal{A}(r) \left(\frac{\Phi(v) - G(v)}{2v} \right)$ with $\mathcal{A}(r) = 3\frac{\sqrt{\pi}}{2} \frac{v^3_{T_{s_0}} \epsilon^{3/2}}{qR_0} \nu_*$ while the expression of the drag term $\mathcal{V}$ is given by $\mathcal{V} = \mathcal{V}_{\|s}(r,v) = -\frac{(v_{G\|} - V_{\|s,\mathrm{coll}})}{v^2_{T_{s,\mathrm{coll}}}} \mathcal{D}_{\|s}(r,v)$. Classically, let us write: $f^n_j \equiv \bar{F}_s(r,\theta,\varphi,v_{G\|,j},t_n)$, where subscript $j \in \{0,\dots,N\}$ refers to the discretised index in parallel velocity space and superscript n refers to the time. For each value of μ_s, each term in Eq. (D.9) reads:

$$B^*_{\|s} \partial_t f \to B^*_{\|s,j} \frac{f^{n+1}_j - f^n_j}{\Delta t} \tag{D.10}$$

$$\begin{aligned} \partial_{v_{G\|}} \left(B^*_{\|s} \mathcal{D} \partial_{v_{G\|}} f \right) \to \frac{1}{2\Delta v^2_{G\|}} \Big\{ & B^*_{\|s,j+\frac{1}{2}} \mathcal{D}_{j+\frac{1}{2}} \left[f^{n+1}_{j+1} - f^{n+1}_j + f^n_{j+1} - f^n_j \right] \\ & - B^*_{\|s,j-\frac{1}{2}} \mathcal{D}_{j-\frac{1}{2}} \left[f^{n+1}_j - f^{n+1}_{j-1} + f^n_j - f^n_{j-1} \right] \Big\} \end{aligned} \tag{D.11}$$

$$\partial_{v_{G\|}} \left(B^*_{\|s} \mathcal{V} f \right) \to \frac{B^*_{\|s,j+1} \mathcal{V}_{j+1} \left[f^{n+1}_{j+1} + f^n_{j+1} \right] - B^*_{\|s,j-1} \mathcal{V}_{j-1} \left[f^{n+1}_{j-1} + f^n_{j-1} \right]}{4\Delta v_{G\|}} \tag{D.12}$$

where we use the fact that $B^*_{\|s}$ is linear in $v_{G\|}$, i.e. the expression $B^*_{\|s,j\pm\frac{1}{2}} = \frac{1}{2} \left(B^*_{\|s,j} + B^*_{\|s,j\pm1} \right)$ is exact. Then, using expressions (D.10), (D.11) and (D.12) in the equation (D.9) gives, for all $j = 0, \cdots, N$:

$$A_j f^{n+1}_{j-1} + B_j f^{n+1}_j + C_j f^{n+1}_{j+1} = -A_j f^n_{j-1} + \left(\frac{2B^*_{\|s,j}}{\Delta t} - B_j \right) f^n_j - C_j f^n_{j+1} \tag{D.13}$$

where the coefficients A_j, B_j and C_j are defined as:

$$A_j = -\frac{\alpha_{j-\frac{1}{2}}}{2\Delta v^2_{G\|}} - \frac{\beta_{j-1}}{4\Delta v_{G\|}} \tag{D.14}$$

$$B_j = \frac{B^*_{\|s,j}}{\Delta t} + \frac{\alpha_{j+\frac{1}{2}} + \alpha_{j-\frac{1}{2}}}{2\Delta v^2_{G\|}} \tag{D.15}$$

$$C_j = -\frac{\alpha_{j+\frac{1}{2}}}{2\Delta v^2_{G\|}} + \frac{\beta_{j+1}}{4\Delta v_{G\|}} \tag{D.16}$$

with

$$\alpha_j = B^*_{\|s,j} \mathcal{D}_j \quad \text{and} \quad \beta_j = B^*_{\|s,j} \mathcal{V}_j \tag{D.17}$$

To compute the terms $\alpha_{-1/2}$, β_{-1}, $\alpha_{N+1/2}$ and β_{N+1} required for A_0, B_0, B_N and C_N calculations, the boundary conditions $\partial^2\mathcal{D}/\partial v^2_{G\|} = 0$ and $\partial^2\mathcal{V}/\partial v^2_{G\|} = 0$ are imposed, i.e :

$$\mathcal{D}_{-\frac{1}{2}} = 2\mathcal{D}_0 - \mathcal{D}_{\frac{1}{2}} \quad \text{and} \quad \mathcal{V}_{-1} = 2\mathcal{V}_0 - \mathcal{V}_1 \tag{D.18}$$

$$\mathcal{D}_{N+\frac{1}{2}} = 2\mathcal{D}_N - \mathcal{D}_{N-\frac{1}{2}} \text{ and } \mathcal{V}_{N+1} = 2\mathcal{V}_N - \mathcal{V}_{N-1} \tag{D.19}$$

and the fact that $B^*_{\|}$ is linear in $v_{G\|}$ is used, i.e [1]:

$$B^*_{\|s,-\frac{1}{2}} = \frac{1}{2}\left(3B^*_{\|s,0} - B^*_{\|s,\frac{1}{2}}\right) \quad ; \quad B^*_{\|s,-1} = 2B^*_{\|s,0} - B^*_{\|s,1} \tag{D.20}$$

$$B^*_{\|s,N+\frac{1}{2}} = \frac{1}{2}\left(3B^*_{\|s,N} - B^*_{\|s,N-\frac{1}{2}}\right) \quad ; \quad B^*_{\|s,N+1} = 2B^*_{\|s,N} - B^*_{\|s,N-1} \tag{D.21}$$

Finally, let us consider that the second derivative of f is vanishing at domain boundaries, i.e

$$f_{-1} = 2f_0 - f_1 \quad \text{and} \quad f_{N+1} = 2f_N - f_{N-1}$$

then

$$(B_0 + 2A_0)f_0^{n+1} + (C_0 - A_0)f_1^{n+1} = \left(\frac{2B^*_{\|s,0}}{\Delta t} - B_0 - 2A_0\right) f_0^n - (C_0 - A_0)f_1^n$$

$$(A_N - C_N)f_{N-1}^{n+1} + (B_N + 2A_N)f_N^{n+1} = -(A_N - C_N)f_{N-1}^n + \left(\frac{2B^*_{\|s,N}}{\Delta t} - B_N - 2C_N\right) f_N^n$$

The system can be rewritten in the compact tridiagonal form:

$$\begin{pmatrix} B_0 + 2A_0 & C_0 - A_0 & & \\ A_1 & B_1 & \ddots & \\ & \ddots & \ddots & C_{N-1} \\ & & A_N - C_N & B_N + 2A_N \end{pmatrix} \begin{pmatrix} f_0^{n+1} \\ \vdots \\ \vdots \\ f_N^{n+1} \end{pmatrix} = \begin{pmatrix} R_0^n \\ \vdots \\ \vdots \\ R_N^n \end{pmatrix} \tag{D.22}$$

and $(R_0^n, \cdots, R_N^n)^t = \mathcal{R}\,(f_0^n, \cdots, f_N^n)^t$ with the matrix $\mathcal{R}$ defined as

$$\mathcal{R} = \begin{pmatrix} -2A_0 - B_0 + \frac{2B^*_{\|s,0}}{\Delta t} & -(C_0 - A_0) & & \\ -A_1 & -B_1 + \frac{2B^*_{\|s,1}}{\Delta t} & \ddots & \\ & \ddots & \ddots & -C_{N-1} \\ & & -(A_N - C_N) & -2C_N - B_N + \frac{2B^*_{\|s,N}}{\Delta t} \end{pmatrix} \tag{D.23}$$

The tridiagonal system (D.22)-(D.23) is solved by using a modified Thomas algorithm (see Appendix K).

[A FAIRE] Actuellement dans le code:

- la première ligne de la matrice $\mathcal{R}$ est $(B_0, C_0, 0, \cdots, 0)$ plutôt que $(B_0 + 2A_0, C_0 - A_0, 0, \cdots, 0)$ d'après (D.23),
- la dernière ligne de la matrice $\mathcal{R}$ est $(0, \cdots, 0, A_N, B_N)$ plutôt que $(0, \cdots, 0, C_N - A_N, B_N + 2A_N)$.

Ce qui est equivalent à $f_{-1}^{n+1} = 0$ et $f_{N+1}^{n+1} = 0$. Jéremie dit qu'il avait fait des tests en RHSonly et c'est ce qui fonctionnait le mieux, i.e Dirichlet pour le temps t^{n+1} et Neumann pour le temps t^n. Résultats à vérifier car en fait $f_{-1}^{n+1} = 0$ et $f_{N+1}^{n+1} = 0$ ne correspondent pas tout à fait à des conditions de Dirichlet.

[1] $$B^*_{\|s,-\frac{1}{2}} = 2B^*_{\|s,0} - B^*_{\|s,\frac{1}{2}} = 2B^*_{\|s,0} - \frac{1}{2}\left(B^*_{\|s,0} + B^*_{\|s,1}\right) = \frac{3}{2}B^*_{\|s,0} - \frac{1}{2}B^*_{\|s,1}$$